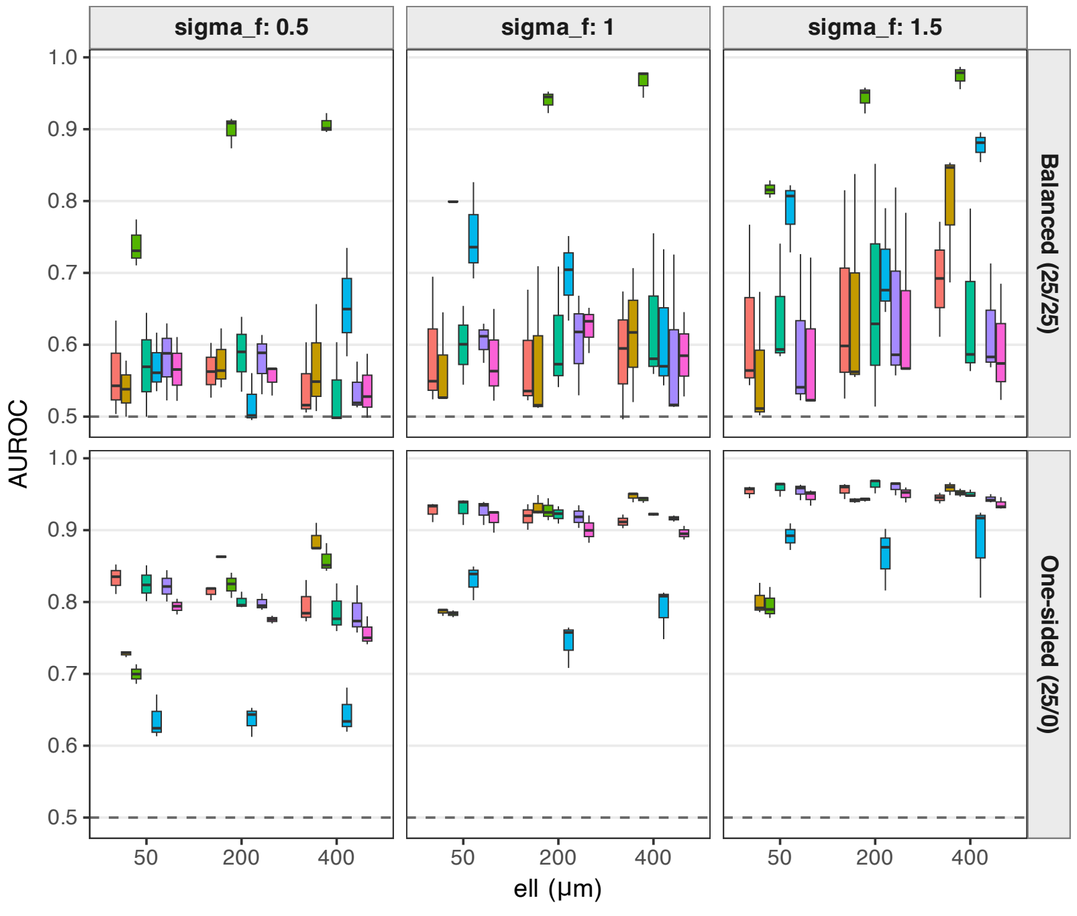
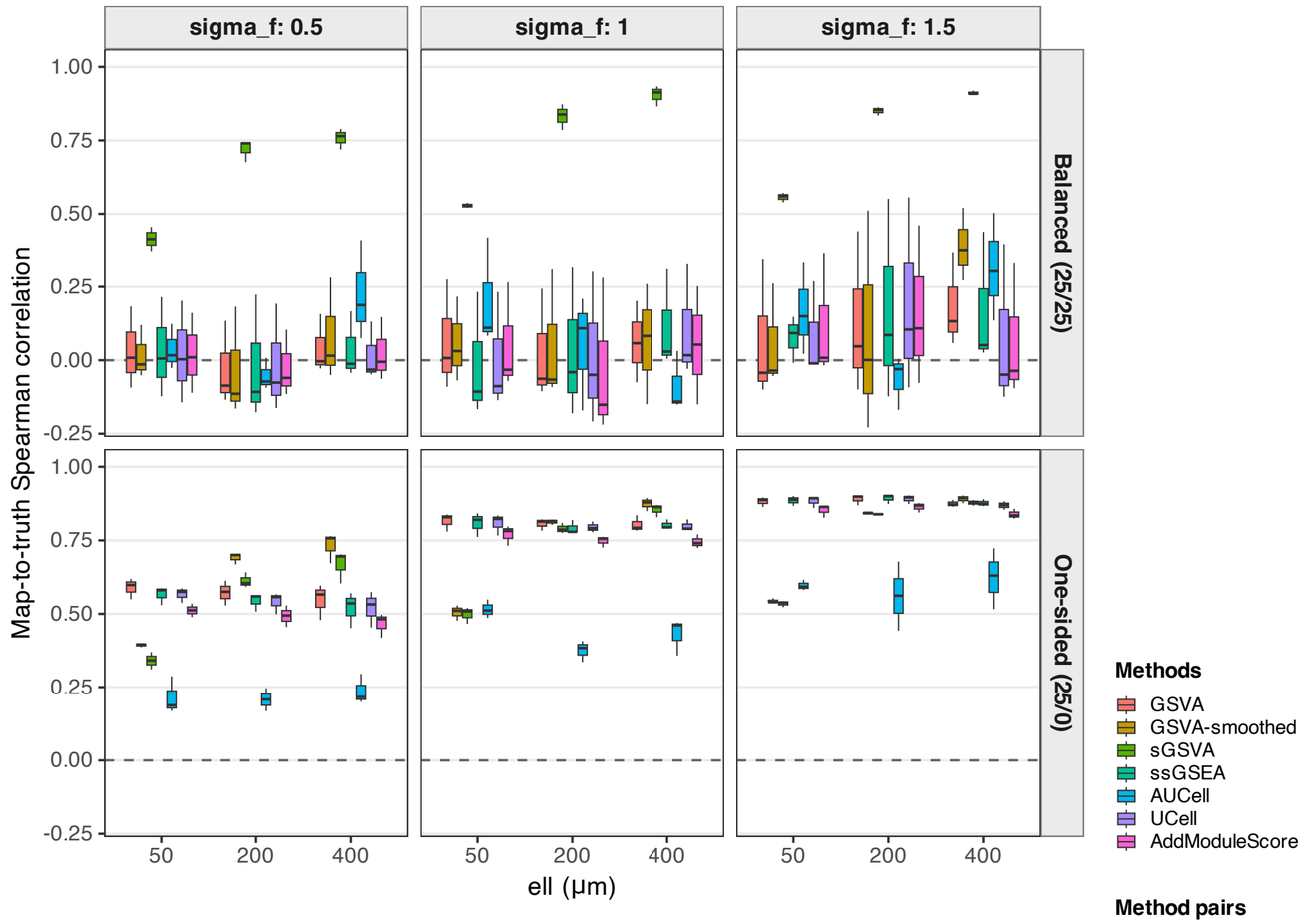
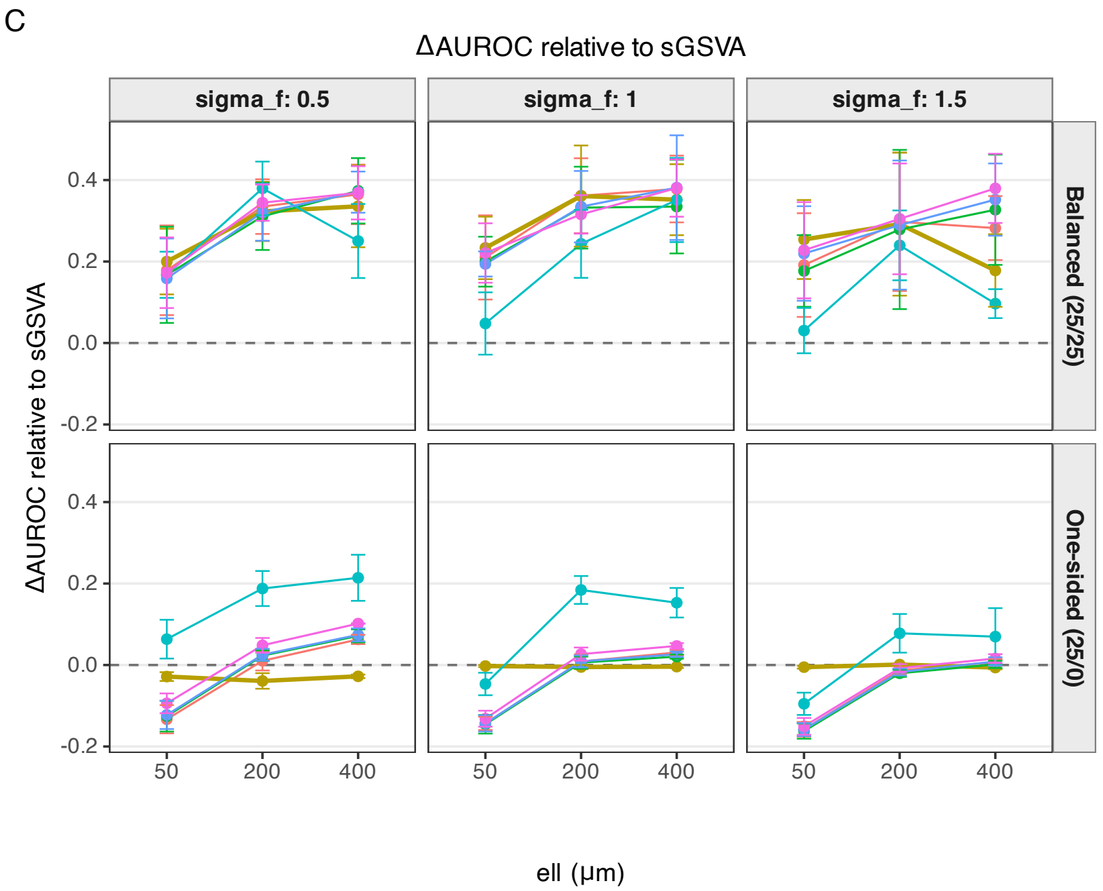
A


Simulation AUROC performance
sigma_f: 0.5
sigma_f: 1
sigma_f: 1.5
Balanced (25/25)
One-sided (25/0)
AUROC
1.0
0.9
0.8
0.7
0.6
0.5
50
200
400
ell (μm)

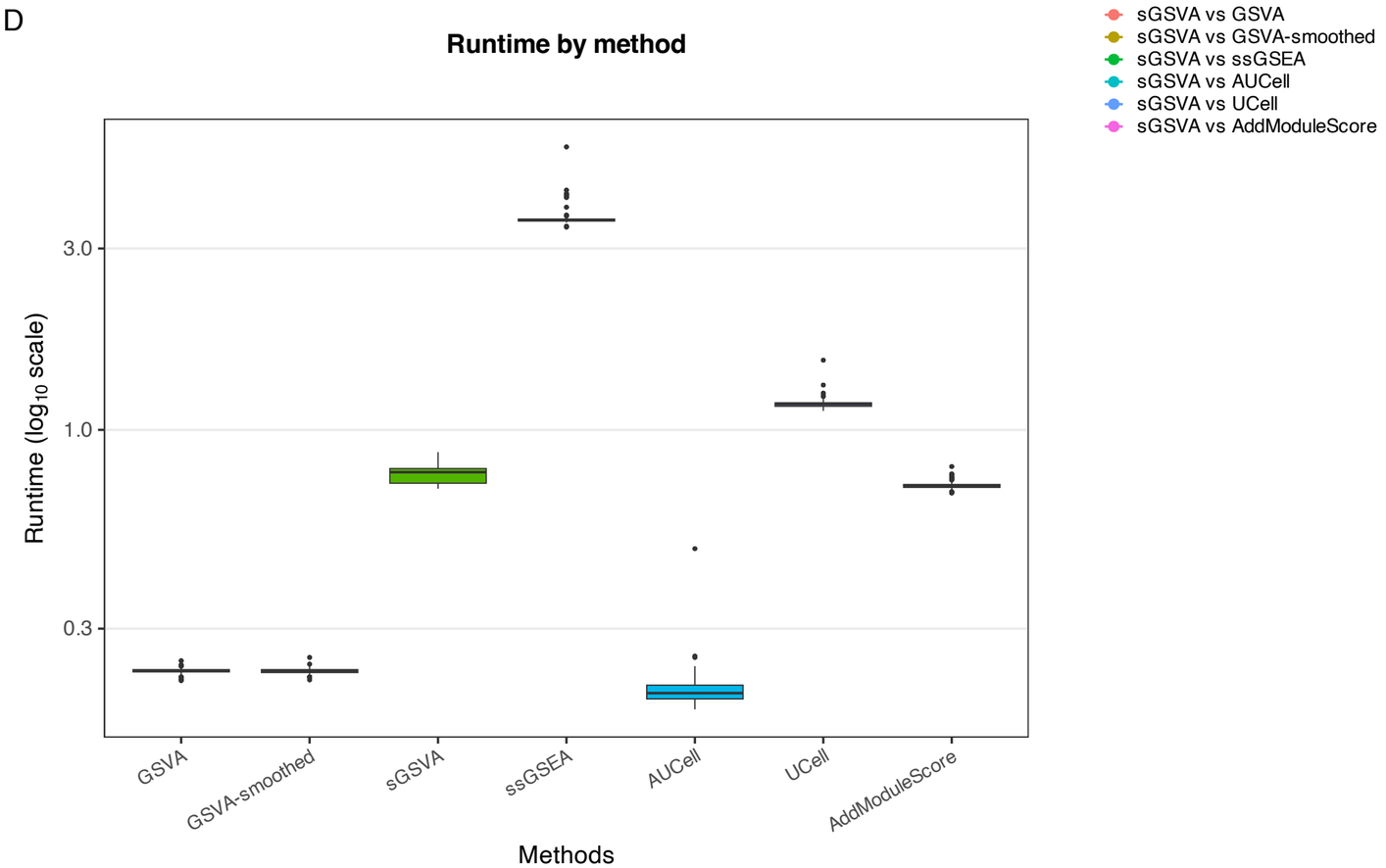
B


Simulation map-to-truth Spearman
sigma_f: 0.5
sigma_f: 1
sigma_f: 1.5
Balanced (25/25)
One-sided (25/0)
Map-to-truth Spearman correlation
1.00
0.75
0.50
0.25
0.00
-0.25
50
200
400
ell (μm)
Methods
GSVA
GSVA-smoothed
sGSVA
ssGSEA
AUCell
UCell
AddModuleScore

C


ΔAUROC relative to sGSVA
sigma_f: 0.5
sigma_f: 1
sigma_f: 1.5
Balanced (25/25)
One-sided (25/0)
ΔAUROC relative to sGSVA
0.4
0.2
0.0
-0.2
50
200
400
ell (μm)

D


Runtime by method
Runtime (log10 scale)
3.0
1.0
0.3
GSVA
GSVA-smoothed
sGSVA
ssGSEA
AUCell
UCell
AddModuleScore
Methods
Method pairs
sGSVA vs GSVA
sGSVA vs GSVA-smoothed
sGSVA vs ssGSEA
sGSVA vs AUCell
sGSVA vs UCell
sGSVA vs AddModuleScore